Day 1 Introduction to Networking

Task 1:- Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.

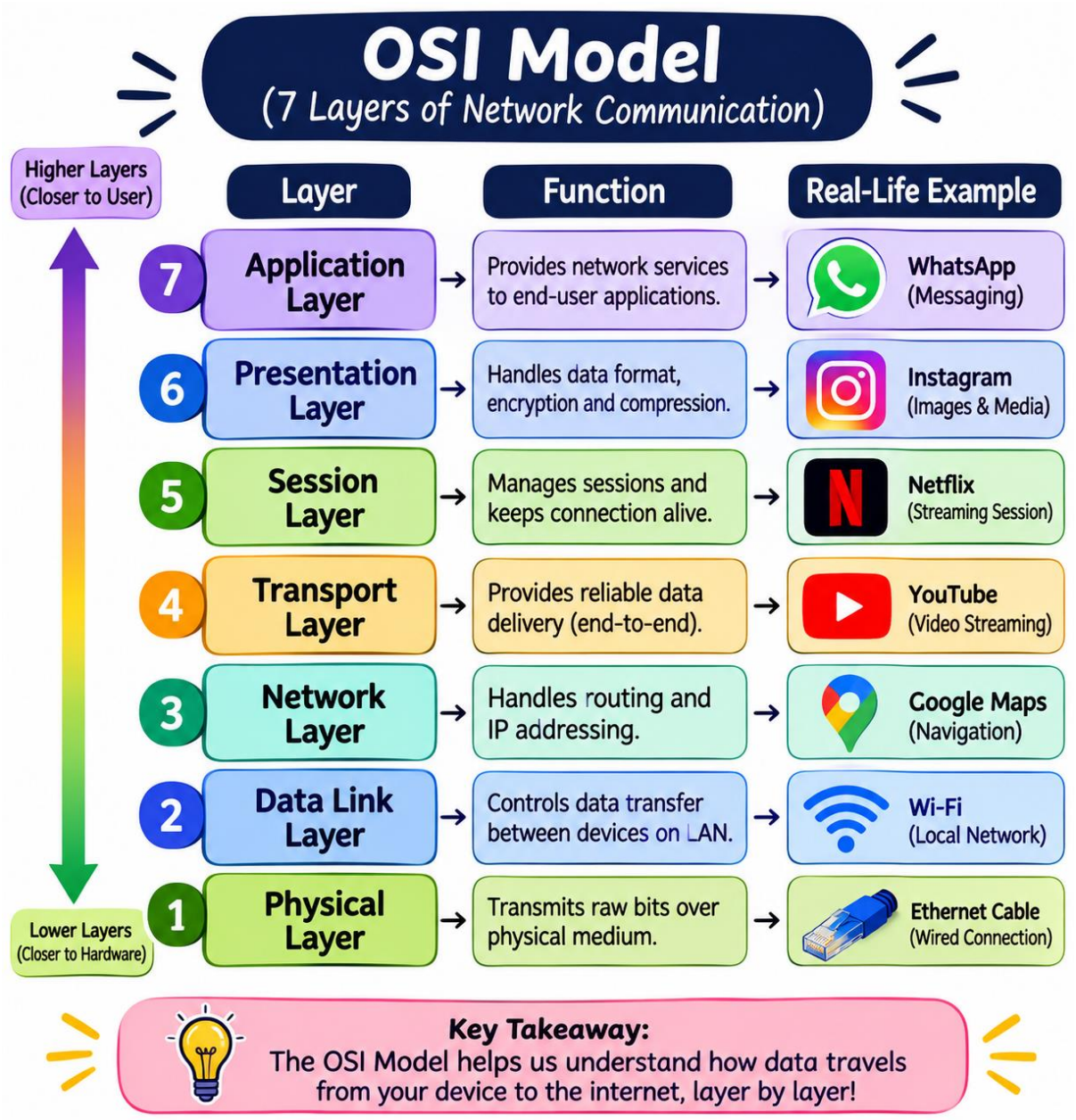
Protocols Noted in Wireshark:

1. DNS
2. TCP
3. TLS
4. QUIC
5. UDP
6. HTTP

Task 2:- Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc.). Use the Protocol column and Info details in Wireshark to help you map protocols to OSI layers.

Packet Protocol	OSI Layer	Function
DNS	Layer 7 – Application	Resolves domain names to IP addresses.
TCP	Layer 4 – Transport	Provides reliable data delivery.
TLS	Layer 6 – Presentation	Encrypts and decrypts data.

Task 3:- Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).



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Task 4:- Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words. Think about how your message is typed, packaged, sent over the internet, and received.

How a WhatsApp Message Travels Through the OSI Model

1. Application Layer (Layer 7): I type a message in WhatsApp and press Send.

2. Transport Layer (Layer 4): Data is transported using protocols such as TCP, helping deliver it reliably.

3. Network Layer (Layer 3): IP addresses help route the data across networks to the destination.

4. Data Link Layer (Layer 2): Data is packaged into frames and sent across the local Wi-Fi network.

5. Physical Layer (Layer 1): The data travels as wireless radio signals from my phone to the Wi-Fi router.

At the receiving end, the data is processed in reverse order until the message appears in the recipient's WhatsApp.